

URVA LATIF

AI / ML ENGINEER • DEEP LEARNING & APPLIED AI SYSTEMS

urvalatif@gmail.com • +92-331-6978231 • linkedin.com/in/urva-latif

PROFILE

AI/ML Engineer with nearly 5 years of experience designing, training, and deploying deep learning models end-to-end — from architecture selection and large-scale dataset pipelines through production deployment and continuous performance monitoring. Proficient in Python and core ML frameworks (PyTorch, TensorFlow, Scikit-learn), with hands-on experience across AWS, GCP, and Azure, and working knowledge of NLP/LLMs, agentic AI systems, and multimodal (vision-language) modelling. Track record of taking models from research to regulated, production-grade deployment, including multiple products that passed external regulatory review (FDA/CE), demonstrating rigor in validation, testing, and documentation that transfers directly to any production ML environment. Comfortable working independently in distributed, remote teams and collaborating cross-functionally with product, engineering, and data stakeholders.

EXPERIENCE

AI Researcher

Jul 2024 – Apr 2026

SaveLife.AI • Texas, USA (Remote)

- Designed, developed, and deployed scalable deep learning models (CNNs, Transformers, 3D U-Net, nnU-Net) across multiple product lines, validating on diverse multi-source datasets to ensure robustness and mitigate bias.
- Owned end-to-end ML pipelines — dataset engineering, preprocessing, model training/optimisation, performance testing, and deployment documentation — across five shipped AI products.
- Built and maintained cloud-based training and deployment workflows (AWS, GCP, Azure) and Dockerised services for reproducible, scalable delivery across environments.
- Implemented explainable AI methods (Grad-CAM, SHAP, LIME) to generate interpretable model outputs supporting end-user decision-making.
- Collaborated cross-functionally with domain experts, data scientists, and engineers to integrate AI models into production workflows, and supported formal external validation and rebuttal processes with technical documentation.

AI Suite — FDA-Cleared & In-Review Imaging Detection Modules

- NeuroICH (CT intracranial haemorrhage detection, FDA 510(k) Cleared): curated large-scale CT datasets and developed a high-sensitivity/specificity detection model with full explainability.
- MammoSightAI (mammogram cancer detection, FDA 510(k) Cleared): built an Otsu-thresholding + ROI-extraction preprocessing pipeline feeding a ConvNeXt-Small classifier.
- CXRDetectAI (chest X-ray abnormality detection, FDA 510(k) Cleared): built and fine-tuned a Faster R-CNN (ResNet50-FPN) detection model; supported validation and rebuttal through to release.
- DenseViewAI (mammographic density classification, FDA submission in review): fine-tuned a MONAI classifier via transfer learning on a limited labelled dataset.
- NeuroLVO (CT perfusion & large vessel occlusion detection, FDA submission in review): automated 4D perfusion-map processing and built expert-consensus annotation workflows.

RadioViewAI — Interactive Imaging Viewer & Measurement Tools

- Integrated an open-source imaging viewer with promptable segmentation (SAM-2, nnInteractive) for 3D volumetric measurement/analysis, and built the Docker-based deployment for the viewer and AI services.

Additional Imaging R&D — Vessel Segmentation, Aneurysm & Multi-Disease Classification

- Built a 3D vessel segmentation pipeline (Frangi filter, MONAI 3D U-Net, VesselFM) and a coarse-to-fine nnU-Net aneurysm-detection workflow classifying presence across 13 anatomical locations.
- Optimised multi-label classification/segmentation models for 18 chest X-ray conditions using attention-based architectures with Grad-CAM explainability.

RadReport & AizaMD — NLP/LLM Contributions

- RadReport: collaborated closely with the NLP team, curating and structuring large-scale medical report datasets to support development of the product's Retrieval-Augmented Generation (RAG) pipeline (Pinecone vector database) for automated report generation.
- AizaMD: experimented with and evaluated transcription models for doctor-patient encounter audio, supporting the ambient documentation pipeline's speech-to-text accuracy.

LLM & Multimodal Prototyping

- Built a RAG-based document Q&A chatbot (vector database + LLM chat interface), a CLIP-style chest X-ray image-and-report retrieval system, and an LLM-driven multimodal patient-summary generator combining imaging findings with clinical notes.

Visiting Lecturer

Nov 2023 – Aug 2024

University of Mianwali • Mianwali, Pakistan

- Delivered lectures and labs; supervised student projects including a semi-supervised active-learning system for sensor-based activity recognition and a PageRank-based social-network analysis tool.

Graphic Designer

Jan 2020 – Sep 2023

PRETECHX IT Solutions • Islamabad, Pakistan

- Created visual identities and layouts for business clients using Adobe Illustrator and Photoshop; managed concurrent projects to deadline using Trello.

Research Associate

Feb 2017 – May 2020

COMSATS University Islamabad • Vehari, Pakistan

- Conducted ML/image-processing research and supervised student projects, including an image-steganography system and a real-time AI-assisted blood-bank matching platform.

Research Engineer

Sep 2015 – May 2016

LMKR Holdings • Texas, USA

- Applied Self-Organizing Maps to cluster seismic attributes and segment geological structures for reservoir detection.

INDEPENDENT PROJECTS

Proposal Automation Agent

- Built a self-directed agentic system to automate drafting of client project proposals, using LLM-based generation with tool-calling agents (LangGraph) to pull context from prior proposals and technical specifications.

EDUCATION

M.S., Computer Science

2017 – 2020

COMSATS University Islamabad, Pakistan — CGPA: 3.99/4.00 | Thesis: An End-to-End Brain Tumor Segmentation System using Multi-Inception-UNET

B.Sc., Electrical Engineering (Telecommunication)

2012 – 2016

National University of Sciences and Technology (NUST), Islamabad — CGPA: 3.74/4.00 | Thesis: Clustering of Features in Seismic Attributes using Self-Organizing Maps (SOM)

SKILLS

Languages & ML Frameworks: Python, PyTorch, TensorFlow, Keras, Scikit-learn, NumPy, Matplotlib, OpenCV, MATLAB

Deep Learning & Modelling: CNNs, Transformers, Attention Models, 3D U-Net, nnU-Net, MONAI, SAM-2, SAM-3, nnInteractive, Transfer Learning, Active Learning, Model Evaluation & Feature Engineering

Cloud & Deployment: AWS (S3, EC2), GCP (Vertex AI), Microsoft Azure, Docker, Cloud-based Model Training & Deployment

Explainability (XAI): Grad-CAM, SHAP, LIME, Model Interpretability

NLP & LLMs: LLM Fine-Tuning, Retrieval-Augmented Generation (RAG), Pinecone, Text Classification, Embeddings & Semantic Search

Agentic AI: LangChain, LangGraph, AutoGen, CrewAI, Custom Tool-Calling Agents, Multi-Agent Orchestration

Multimodal AI: Vision-Language Models (CLIP/LLaVA-style), Image + Text/Report Pipelines

Data & Domain Applications: Large-scale Dataset Pipeline Engineering, Performance Testing, Regulatory Documentation (FDA 510(k), CE), Medical Imaging (CT, CTA, X-ray, Mammography, MRI)

Tools & Collaboration: Git, GitHub, Bitbucket, Jira, Trello, Slack, Notion, ITK-SNAP, NIfTI, 3D Slicer, Cursor, Claude Code

SELECTED PUBLICATIONS

- Latif U, Shahid AR, Raza B, Ziauddin S, Khan MA. An end-to-end brain tumour segmentation system using multi-inception-UNET. Int J Imaging Syst Technol. 2021. DOI: 10.1002/ima.22585

ADDITIONAL

- Scholarships: HEC-funded MS thesis (Grant #2(1064)), 6-semester merit scholarship. Training: Kaggle (AI Ethics, ML Explainability), Huawei Graduate Training Program, FAB Internship.